Temporal Data Subset Stability (DSS) for Artifact Reduction in Photoacoustic Computed Tomography

Soheil Hakakzadeh[a*], Amirreza Jodeiry[b], Zahra Kavehvash [c], Mohammad Mehrmohammadi [d,e,f]

[a] Department of Electrical and Computer Engineering, University of Tehran, Tehran, Iran
[b,c] Department of Electrical Engineering, Sharif University of Technology, Tehran, Iran
[d] Department of Imaging Science, University of Rochester Medical Center, NY, USA
[e] Department of Electrical and Computer Engineering, University of Rochester, NY, USA
[f] Department of Biomedical Engineering, University of Rochester, NY, USA

ABSTRACT

Photoacoustic computed tomography (PACT) often suffers from sparse-sampling streak artifacts that severely degrade image contrast. While recent temporal weighting algorithms successfully mitigate background noise in dynamic imaging, they often suppress weak microvascular signals. We propose a Temporal Data Subset Stability (T-DSS) framework that calculates pixel-wise consistency across temporal frame lags to distinguish physically stable absorbers from fluctuating artifacts. Validated on an in vivo 600-frame mouse dataset, the non-iterative T-DSS method successfully eliminates background artifacts without missing critical anatomical data. Quantitative evaluation demonstrates that T-DSS achieves a superior generalized contrast-to-noise ratio (gCNR) of 0.74, outperforming both the conventional delay-and-sum baseline (0.65) and the state-of-the-art DARPIC method (0.45).

Keywords: Photoacoustic computed tomography, Artifact reduction, Temporal stability, Multi-frame imaging.

1. INTRODUCTION

Photoacoustic computed tomography (PACT) is an emerging biomedical imaging modality combining high optical contrast with acoustic spatial resolution [1]. However, sparse detector sampling—often required due to hardware constraints and computational burdens—inevitably generates streak artifacts that severely degrade image contrast and obscure real absorbers [2], [3]. To suppress these artifacts, numerous single-frame reconstruction strategies have been proposed, ranging from adaptive weighting [4] and coherence factors [5] to model-based iterations [6] and deep learning [7]. In the spatial domain, we previously introduced the random detection points (RDP) framework, which successfully exploited spatial reconstruction inconsistencies to differentiate artifacts from true absorbers [8]. Inspired by this concept, Zhang and his coworkers recently developed the regularized iteration method with structural prior (RISP) method, incorporating the randomized structural prior into an iterative regularization framework [9]. Furthermore, we subsequently introduced the spatial Detector Subset Stability (DSS) framework to rigorously evaluate pixel-wise structural reliability across varying spatial subarrays [10].

In dynamic in vivo imaging, consecutive multi-frame acquisitions provide a vital temporal dimension. Recently, algorithms like DARPIC have demonstrated that streak artifacts fluctuate significantly with micro-target motion across frames, whereas true anatomical features remain temporally stable [11]. While DARPIC successfully mitigates artifacts, its reliance on non-linear transformations inadvertently over-penalizes slight intensity variations, resulting in the suppression of fine microvascular structures. To overcome these limitations, we extend our spatial DSS approach [9] into the temporal domain by introducing Temporal Data Subset Stability (T-DSS). By calculating a pixel-wise temporal similarity metric across registered temporal data subsets, T-DSS evaluates stability strictly in the temporal domain. Validated using an open-access 600-frame in vivo dataset [12], T-DSS successfully isolates stable absorbers and eliminates artifacts while rigorously preserving the fine structural data points omitted by existing methods.

2. METHODOLOGY

2.1 Frame-wise Reconstruction and Subsampling Strategy

As illustrated in the block diagram (Figure 1), the proposed T-DSS framework processes a dynamic sequence of raw multi-frame photoacoustic data. First, each frame is individually reconstructed using the conventional DAS algorithm. To ensure temporal alignment, motion registration is applied to the frame-wise reconstructions. These registered frames are then summed to create a single, high-SNR composite baseline image, denoted as the Summed DAS Image.

*soheil.hakakzadeh@ut.ac.ir

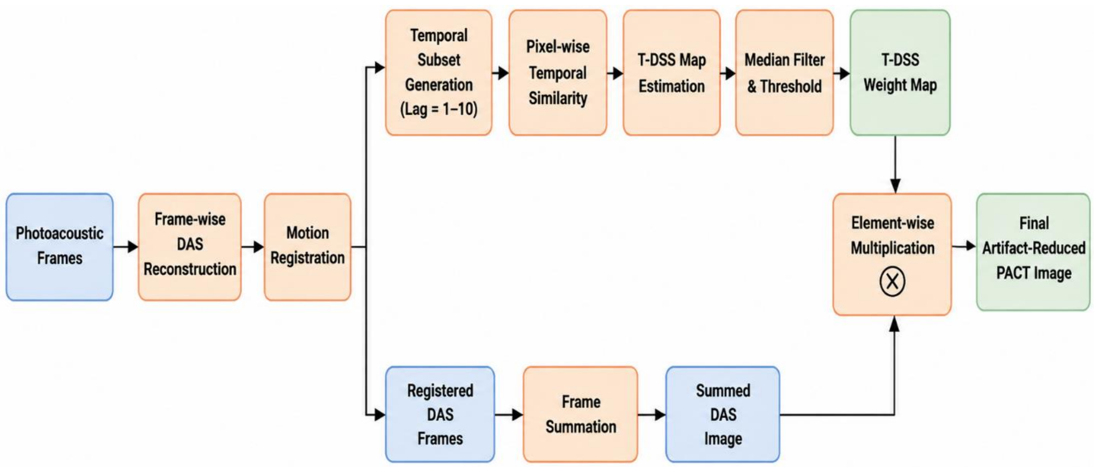

Figure 1.Workflow diagram of the proposed temporal DSS method.

Concurrently, to exploit the temporally fluctuating behavior of streak artifacts, we adapt our previously established inconsistency analysis framework [8] to operate strictly on the temporal axis. We partition the registered dynamic sequence into distinct temporal data subsets (e.g., frame pairs separated by varying temporal lags of 1–10). Because streak artifacts fluctuate dynamically due to physiological micro-motion across different temporal frames, the artifact profiles differ significantly among these temporal subsets, whereas the locations and intensities of true absorbers remain geometrically and temporally stable.

2.2 Pixel-wise Temporal Similarity and Coefficient of Variation (CV) Definition

To distinguish stable PA absorber regions from unstable artifact-prone regions, we analyze pixel-level consistency across the temporal data sequence. Let $F(r,t)$ denote the reconstructed photoacoustic intensity at spatial pixel r for the registered frame at temporal index t.
In this multi-frame framework, the normalized temporal similarity is computed between corresponding pixels in pairs of frames separated by a temporal lag τ. For a frame at time t and its delayed counterpart at time $t+\tau$, the pixel-wise temporal similarity $S_{t,\tau}(r)$ is defined as:

$$S_{t,\tau}(r) = \frac{F(r,t)\cdot F(r,t+\tau)}{\max_{r}[F(r,t)\cdot F(r,t+\tau)]}. \tag{1}$$

To quantify the variation in pixel-wise temporal correlation, we define the CV. Assuming K total evaluated frame pairs across the temporal subsets, we first compute the mean correlation across all frame pairs at pixel r, formulated as $\mu_S(r) = \frac{1}{K}\sum_{t,\tau} S_{t,\tau}(r)$. Next, we compute the standard deviation of the pixel-wise correlations, given by: $\sigma_S(r) = \sqrt{\frac{1}{K}\sum_{t,\tau}[S_{t,\tau}(r) - \mu_S(r)]^2}$ .The coefficient of variation is then defined as below:

$$CV(r) = \frac{\mu_S(r)}{\sigma_S(r)}. \tag{2}$$

High CV values indicate stable absorber regions with consistently strong correlation, whereas low CV values correspond to unstable artifact-dominated regions.

2.3 T-DSS Weight Map Generation and Application

To suppress isolated outliers caused by unstable artifact responses, a spatial median filter is applied to the CV map. To identify reliable absorber regions, a percentile-based thresholding approach is applied to segment the processed map, generating the final binary T-DSS Weight Map, denoted as $\hat{S}(r)$.
Once the map $\hat{S}(r)$ is computed, it is directly applied to the Summed DAS Image [denoted as $p_0^{full}(r)$]. The final artifact-suppressed image is obtained by pixel-wise multiplication as below:

$$p_0^{final}(r) = \hat{S}(r)\cdot p_0^{full}(r). \tag{3}$$

Here, $\hat{S}(r) \in \{0,1\}$ acts as a spatial selector: it retains values in stable (absorber) regions and suppresses values in temporally unstable (artifact-prone) regions. By using this stability-guided strategy strictly across the temporal frame sequence, the proposed framework yields an overall better, artifact-free final single image from the dynamic multi-frame data.

3. RESULTS

3.1 In Vivo Dataset and Experimental Setup

To validate the proposed T-DSS framework and provide a direct comparison with state-of-the-art temporal methods, we utilized an open-access *in vivo* photoacoustic dataset provided by Ma et al. [11], [12]. The dataset consists of 600 consecutive RF data frames acquired from the upper abdominal cavity of an anesthetized mouse. Data was captured using a 256-element full-ring transducer array operating at a 20 Hz frame rate. This dynamic sequence captures the micro-motion necessary to induce variations in streak artifacts for temporal consistency analysis.

3.2 Qualitative Evaluation

Figure 2 compares the reconstruction performance of the conventional DAS algorithm, the DARPIC method, and the proposed T-DSS framework applied to the summed 600-frame sequence. The standard DAS reconstruction [Fig. 2(a1) and magnified in 2(a2)] retains the raw photoacoustic signal but is heavily corrupted by severe streak artifacts and background acoustic noise, which obscure fine anatomical details.

While the DARPIC algorithm [Fig. 2(b1) and 2(b2)] successfully suppresses the background artifacts, visual inspection reveals that it inadvertently attenuates weak photoacoustic signals from genuine structures. As indicated by the gray arrows in the magnified view [Fig. 2(b2)], several fine microvascular branches and boundary details are missing, disjointed, or significantly dimmed.

In contrast, the proposed T-DSS framework [Fig. 2(c1) and 2(c2)] achieves a clean, artifact-free background while meticulously preserving the continuous morphology and intensity of the microvasculature. By enforcing stability across the temporal frame sequence, T-DSS prevents the over-penalization of slight intensity variations. As pointed out by the gray arrows in Fig. 2(c2), T-DSS successfully recovers and preserves critical microvascular data points that were entirely missed or degraded by the DARPIC method.

3.3 Quantitative Evaluation

To quantitatively assess reconstruction quality, we evaluated the generalized contrast-to-noise ratio (gCNR). The conventional DAS composite yields a baseline gCNR of 0.65 due to severe background streak artifacts. While the DARPIC algorithm visually clears these background regions, its aggressive non-linear weighting inadvertently over-penalizes weak microvascular signals. This erasure of true structural intensity degrades the overall image contrast,

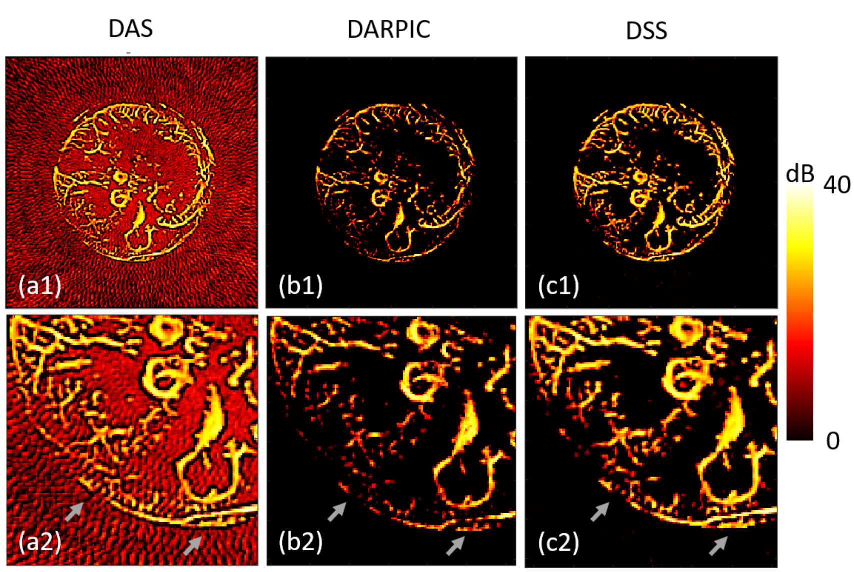

Figure 2. Comparison of in vivo artifact suppression. (a1) DAS composite, (b1) DARPIC, (c1) Proposed T-DSS. (a2–c2) Magnified regions. Gray arrows highlight subtle microvascular details and boundaries that are inadvertently suppressed by DARPIC but successfully preserved by T-DSS.

dropping its gCNR to 0.45. In contrast, the proposed T-DSS framework effectively eliminates background artifacts while meticulously preserving true absorber intensities. Consequently, T-DSS achieves the highest gCNR of 0.74, mathematically demonstrating superior background artifact suppression without sacrificing critical anatomical data.

4. DISCUSSION AND CONCLUSION

In this work, we introduced the Temporal Data Subset Stability (T-DSS) framework to address sparse-sampling streak artifacts in multi-frame circular PACT. By calculating pixel-wise consistency across temporal frame lags, T-DSS effectively distinguishes physically stable true absorbers from fluctuating artifacts.
Validated on an in vivo mouse dataset, T-DSS overcomes the fundamental limitations of existing temporal weighting approaches like DARPIC. While DARPIC successfully reduces background noise, its reliance on phase-aware clustering and non-linear transformations inadvertently over-penalizes slight intensity variations, leading to the loss of weak photoacoustic signals. In contrast, our temporal approach achieves comparable or superior artifact elimination without sacrificing data. Visual and quantitative results demonstrate that T-DSS meticulously preserves delicate microvascular structures, boundary details, and true absorber intensities that other methods suppress. Ultimately, T-DSS provides a highly reliable, non-iterative solution for generating artifact-free, high-fidelity composite PACT images, paving the way for improved dynamic clinical imaging.

Disclosure of prior or concurrent submissions: This work is original and has not been previously published, presented, or concurrently submitted elsewhere.

REFERENCES

[1] L. V. Wang and S. Hu, "Photoacoustic Tomography: In Vivo Imaging from Organelles to Organs," *Science*, vol. 335, no. 6075, Art. no. 6075, Mar. 2012, doi: 10.1126/science.1216210.

[2] M. Xu and L. V. Wang, "Universal back-projection algorithm for photoacoustic computed tomography," *Phys. Rev. E*, vol. 71, no. 1, p. 016706, Jan. 2005, doi: 10.1103/PhysRevE.71.016706.

[3] A. Rosenthal, V. Ntziachristos, and D. Razansky, "Acoustic Inversion in Optoacoustic Tomography: A Review," *Curr. Med. Imaging Rev.*, vol. 9, no. 4, Art. no. 4, Jan. 2014, doi: 10.2174/15734056113096660006.

[4] G. Matrone, A. S. Savoia, G. Caliano, and G. Magenes, "The Delay Multiply and Sum Beamforming Algorithm in Ultrasound B-Mode Medical Imaging," *IEEE Trans. Med. Imaging*, vol. 34, no. 4, Art. no. 4, Apr. 2015, doi: 10.1109/TMI.2014.2371235.

[5] R. A. Mukaddim and T. Varghese, "Spatiotemporal Coherence Weighting for *In Vivo* Cardiac Photoacoustic Image Beamformation," *IEEE Trans. Ultrason. Ferroelectr. Freq. Control*, vol. 68, no. 3, pp. 586–598, Mar. 2021, doi: 10.1109/TUFFC.2020.3016900.

[6] X. L. Deán-Ben and D. Razansky, "A practical guide for model-based reconstruction in optoacoustic imaging," *Front. Phys.*, vol. 10, p. 1028258, Nov. 2022, doi: 10.3389/fphy.2022.1028258.

[7] S. Li *et al.*, "Zero-Shot Artifact2Artifact: Self-incentive artifact removal for photoacoustic imaging," *Photoacoustics*, vol. 43, p. 100723, Jun. 2025, doi: 10.1016/j.pacs.2025.100723.

[8] S. Hakakzadeh, P. Rajendran, Z. Kavehvash, and M. Pramanik, "Enhancing image quality in circular-view photoacoustic tomography using randomized detection points," *J. Phys. Photonics*, vol. 7, no. 1, p. 015006, Jan. 2025, doi: 10.1088/2515-7647/ad9b83.

[9] Y. Zhang *et al.*, "Iterative optimization algorithm with structural prior for artifacts removal of photoacoustic imaging," *Photoacoustics*, vol. 44, p. 100726, Aug. 2025, doi: 10.1016/j.pacs.2025.100726.

[10] "Detector-Subset Stability (DDS)-Guided Artifact Suppression in Circular Photoacoustic Computed Tomography".

[11] Z. Ma *et al.*, "Multi-frame pixel intensity consistency-based artifact reduction for photoacoustic computed tomography," *Biomed. Opt. Express*, vol. 17, no. 5, p. 2660, May 2026, doi: 10.1364/BOE.593136.

[12] Z. Ma, "Darpic-dynamic-artifact-reduction-via-pixel-intensity-consistency*," GitHub, 2026, https://github.com/Yusheng349/DARPIC-Dynamic-Artifact-Reduction-via-Pixel-Intensity-Consistency.*